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Project PROPOSAL

BUP Social — A University-Centered Social, Academic & Campus Services Platform (BUPSoC)

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Abstract

BUP Social is a university-focused web application that brings everyday student communication and selected campus services into one organized environment. It connects social interaction, academic groups, routines, events, clubs, and transportation through verified BUP accounts and an academic-aware structure.

The platform provides profiles, social feeds, academic communities, club and event discovery, messaging, notifications, a personalized Routine Hub, transportation information, RideShare, Travel Partner discovery, Lost & Found, search, reporting, and administrative controls. It uses a modular client-server architecture with a relational database, REST APIs, authentication, and role-based authorization.

The proposed platform is intended to reduce information fragmentation by organizing communication and services according to BUP-specific academic and campus structures. Students will be able to access relevant communities, events, routines, transportation opportunities, and campus information from a single platform. Faculty members and verified clubs can publish authorized information and activities, while moderators and administrators can manage verification, reports, routines, and platform governance.

Keywords: BUP Social, University Social Platform, Academic Services, Campus Services, Student Engagement, University Communication

1 Introduction

University communication plays an important role in keeping students, faculty members, clubs, and organizations connected with academic and campus activities. However, university communication is often fragmented across messaging groups, social-media pages, shared files, notices, and informal conversations. Although these channels are useful individually, they are not organized around the academic structure of a university. As a result, information related to courses, departments, batches, sections, clubs, events, and other campus activities can be distributed across different platforms.

A student may therefore need to search several places for a class update, event announcement, club activity, routine change, academic information, or transportation opportunity. This fragmented communication process can make relevant information difficult to discover and may require students to depend on multiple groups or informal sources. Similarly, faculty members and verified clubs may need separate channels to communicate with appropriate student communities.

BUP Social proposes a single university-centered digital environment that organizes communication and selected campus services around faculty, department, program, batch, section, course, club, and event structures. The platform provides students with an organized space to discover relevant information, participate in academic and social communities, communicate with other users, and access selected campus-oriented services. Faculty members, clubs, and university administrators can also use the platform according to their respective roles.

The platform is designed around the communication and coordination needs of the BUP community while complementing, rather than replacing, general-purpose social networks.

2 Problem Statement

The current communication process creates five related problems: information is distributed across unrelated channels; important updates can be buried or become outdated; academic and extracurricular activities are difficult to discover systematically; transportation coordination relies on informal communication; and external groups lack a common BUP-specific governance model.

- **Information fragmentation:** students must monitor multiple groups and platforms for different categories of information.
- **Weak discoverability:** clubs, events, academic communities, and student opportunities may not reach the students most interested in them.
- **Routine inconsistency:** schedules may circulate as images, PDFs, or messages without a clear indication of the latest version.
- **Transportation coordination:** students lack a dedicated, university-centered mechanism for sharing rides or finding compatible commuting partners.
- **Moderation and trust:** external groups do not provide a common BUP-specific verification and governance model.

Proposed response: one university-centered platform that combines structured communication and campus services while keeping access, verification, and administration under clear role-based controls.

3 Objectives

1. Design and develop a centralized web platform for BUP student communication and campus-oriented services.

2. Organize users and communities according to relevant BUP academic structures.

3. Provide a reliable Routine Hub with personalized schedules and authorized change notifications.

4. Improve discovery of clubs, events, academic communities, and student opportunities.

5. Support student transportation coordination through RideShare and Travel Partner features.

6. Provide private messaging, notifications, Lost & Found, search, reporting, and moderation.

7. Implement authentication, role-based authorization, privacy controls, and secure data handling.

8. Deliver a responsive and maintainable prototype suitable for further institutional development.

4 Scope of the Project

4.1 In Scope

- User registration, authentication, profiles, privacy settings, and account management.
- Social feed, academic communities, verified clubs/organizations, events, and messaging.
- Routine Hub with personalized schedules and authorized routine updates.
- Transportation, RideShare, Travel Partner, Lost & Found, search, moderation, dashboards, and reports.

4.2 Out of Scope / Future Enhancement

- Native Android/iOS applications in the initial academic prototype.
- Live bus/GPS tracking without appropriate institutional or transport infrastructure.
- Direct integration with protected BUP academic/ERP systems without institutional authorization.
- Advanced AI recommendation, automated moderation, and intelligent route matching.

4.3 Target Users

- The primary target users of BUP Social are BUP students, faculty members, verified clubs and organizations, moderators, and system administrators. Students will use the platform for academic and social communication, routine management, event discovery, transportation coordination, and other campus-oriented services. Faculty members will use the platform to communicate authorized academic information. Verified clubs and organizations will manage their activities and events, while moderators and administrators will manage verification, moderation, routines, reports, and platform governance.

5 Stakeholders

Stakeholder	Primary Interaction	Value Provided
Students	Use profiles, feed, communities, routines, events, messaging, Lost & Found and transportation services.	One place to find relevant information and coordinate campus activities.
Faculty Members	Participate in academic communities and publish authorized academic information or notices.	More structured communication with appropriate student groups.
Clubs & Organizations	Maintain verified pages, publish activities, create events and communicate with members/followers.	Better visibility, recruitment and event participation.
Moderators	Review reports, handle inappropriate content and apply community rules.	Consistent platform governance and safer interactions.
Administrators	Manage users, verification, academic data, routines, clubs, moderation and reports.	Centralized oversight and auditable administration.
Project Team	Analyze, design, implement, test, document and deploy the prototype.	Demonstrates practical software engineering skills across the full lifecycle.

5.1 Requirement Elicitation

- Initial requirements for BUP Social were identified through discussions with relevant BUP students and stakeholders regarding existing communication, academic coordination, event discovery, transportation, and campus-service challenges. The discussions focused on identifying frequently used communication channels, difficulties in finding relevant information, routine and event updates, transportation coordination, and moderation requirements. The findings were used to identify the major system features and to define the preliminary functional and non-functional requirements of the proposed platform.

5.1.1 Key Findings

1. Communication difficulties
2. Academic coordination issues
3. Difficulty finding relevant information
4. Routine & event update challenges
5. Transportation coordination needs
6. Moderation requirements

6 Project Requirements

Requirements are written as testable statements so that implementation and testing can be traced back to defined expected behavior. Functional requirements describe capabilities; non-functional requirements define quality expectations.

6.1 Functional Requirements

ID	Functional Requirement
FR-01	The system shall authenticate registered users before granting access to protected resources.
FR-02	The system shall support verification and role assignment for eligible BUP users and organizations.
FR-03	The system shall allow users to maintain profiles, academic information, and privacy preferences.
FR-04	The system shall allow authorized users to create, edit, remove, react to, comment on, and report posts.
FR-05	The system shall support academic communities and user discovery using relevant BUP attributes.
FR-06	The system shall allow verified clubs/organizations to manage pages, posts, and events.
FR-07	The system shall allow students to discover and register/express interest in events.
FR-08	The system shall provide one-to-one and group messaging with notification support.
FR-09	The system shall generate a personalized routine view from the user's academic profile.
FR-10	The system shall allow authorized personnel to publish routine changes and notify affected users.
FR-11	The system shall display transportation information and support RideShare offers and requests.
FR-12	The system shall match RideShare requests using route, time, date, and seat constraints where applicable.
FR-13	The system shall allow students to discover Travel Partners using approximate areas rather than public exact addresses.
FR-14	The system shall provide Lost & Found records with status and reporting capabilities.
FR-15	The system shall provide search across major platform entities such as users, groups, events and posts.
FR-16	The system shall provide administrative moderation, user management, reporting, and basic system statistics.

6.2 Non-functional Requirements

Quality Attribute	Target / Constraint
Performance	Normal page/API interactions should target approximately ≤2 seconds under expected prototype load.
Security	Protected endpoints shall enforce authentication and server-side authorization; passwords shall be securely hashed.
Privacy	Exact home addresses shall not be publicly displayed; sensitive ride/contact details shall be limited to appropriate participants.
Reliability	Important records shall be persisted consistently, with backup procedures for the deployed prototype.
Scalability	Modules and database relationships shall be structured so additional users and records can be accommodated without redesigning the entire system.
Usability	The interface shall remain understandable and responsive on common desktop and mobile browsers.
Maintainability	The codebase shall use modular components, documented APIs/data models, Git version control, and separation of concerns.
Auditability	Important administrative actions should record the actor, timestamp, action type, and relevant status/reason.

7 Proposed Methodology

The project will use an iterative Agile-inspired workflow rather than treating the system as one large implementation task. Requirements and interface decisions will be validated early; development will proceed by feature groups; and each cycle will end with an integrated, testable build. Git-based version control and issue tracking will be used to coordinate work between team members.

7.1 Development Approach

- Requirements & modeling: identify actors, use cases, data entities, constraints, and acceptance criteria.
- UI/UX prototyping: prepare low- and high-fidelity screens before implementing complex interactions.
- Foundation: establish authentication, database schema, API conventions, error handling, and authorization.
- Feature increments: implement social/academic, clubs/events, routines, messaging, and transportation modules in successive increments.
- Integration: connect modules through common APIs, notifications, permissions, and shared data.
- Verification: perform functional, integration, security, usability, and regression testing.
- Deployment & documentation: deploy the prototype, record setup/configuration, and prepare the final demonstration.

The sequence keeps validation close to implementation: modeling and interface work establish the intended behavior, the foundation provides the shared platform services, feature increments build the major modules, and integration and verification confirm that the complete prototype works together before deployment and final documentation.

7.1.1 Sprint and Team Management

The development will be organized into **two-week iterative development cycles (sprints)**. Each sprint will focus on a specific group of related features and will produce an integrated, testable build. Tasks will be managed using a shared project board with the stages **Backlog, In Progress, Review, Testing, and Done**. Team members will be assigned tasks according to the functional modules and project responsibilities.

The team will conduct **weekly meetings** to review completed work, discuss technical or integration issues, redistribute tasks when necessary, and plan the next development cycle. Git and GitHub will be used for source-code management through feature branches, with changes reviewed and merged into the main branch after testing.

This iterative approach is suitable for BUP Social because the system contains multiple interconnected modules, including social communication, academic communities, clubs and events, routine management, messaging, and transportation. Developing and validating these modules progressively will allow the team to identify integration, usability, and requirement issues before the final deployment.

7.2 Technology Stack

Layer	Proposed Technology	Reason for Selection
Frontend	**React + TypeScript + responsive CSS**	Component-based UI, reusable modules, type safety and responsive layouts.
Backend	**Node.js + Express.js**	Lightweight REST API layer suitable for modular web services and asynchronous operations.
Database	**MySQL**	Relational model fits users, academic structures, posts, events, routines and ride records.
Authentication	**JWT / secure session strategy + bcrypt**	Supports authenticated API access, password protection and role-based authorization.
Real-time	**WebSocket / Socket.IO (where required)**	Suitable for chat and timely notifications without coupling every feature to a persistent socket connection.
API / Testing	**REST, Postman**	Clear service boundary and practical API verification during development.
Design / Collaboration	**Figma, Git, GitHub**	Supports interface prototyping, source control, branching and team collaboration.
Deployment	**Vercel + Render**	Supports convenient deployment of the frontend and backend and integrates well with GitHub-based development workflows.

Architecture Description

The proposed system follows a **three-layer client-server architecture** consisting of the presentation layer, application layer, and data layer. The **React and TypeScript frontend** provides the user interface and enables students, faculty members, clubs, and administrators to interact with the platform through a responsive web interface. It communicates with the backend using **REST APIs** for sending user requests and receiving processed results. The **Node.js and Express.js backend** acts as the application layer and manages authentication, authorization, business logic, input validation, and communication between different modules of the system. It handles major platform functions such as social feeds and communities, academic information, routine management, clubs and events, messaging, notifications, RideShare, Travel Partner, Lost & Found, and reporting and moderation.
The **MySQL database** serves as the data layer and stores and manages user profiles, academic information, community and post data, club and event information, routines, messages, transportation-related information, reports, and other required records. When a user performs an action, the request is first received by the frontend and forwarded to the backend through the appropriate REST API. The backend validates and processes the request, performs the required business operations, and communicates with the MySQL database when data needs to be stored, retrieved, or updated. The processed response is then returned to the frontend and presented to the user. This separation of **presentation, application logic, and data management** supports modular development, easier testing, maintainability, scalability, and future enhancement of the BUP Social platform.

7.3 System Architecture

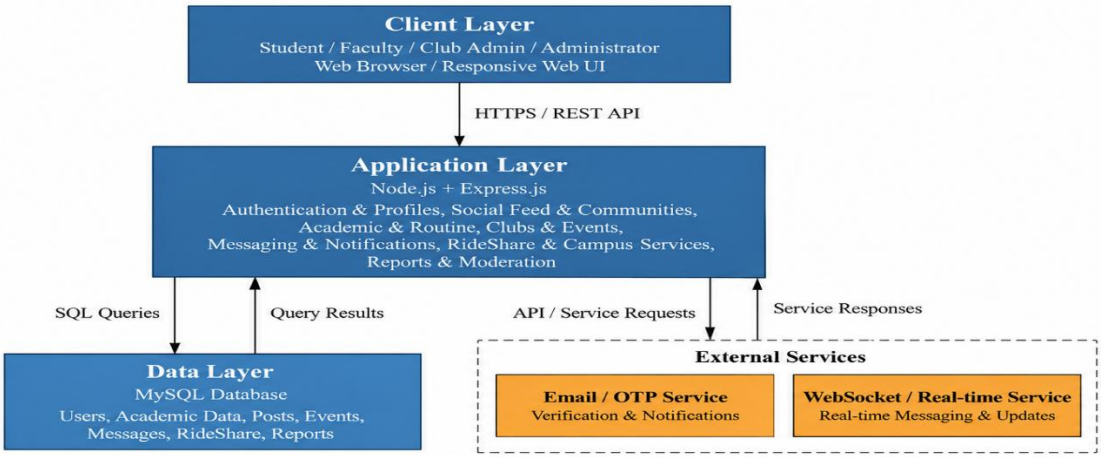


Figure: High-Level System Architecture of BUP Social

7.4 Context View

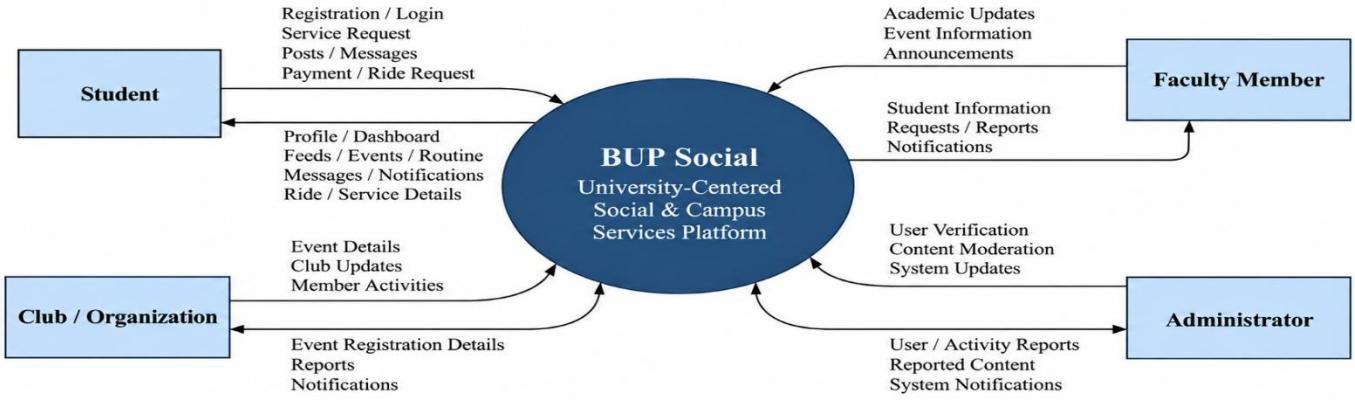


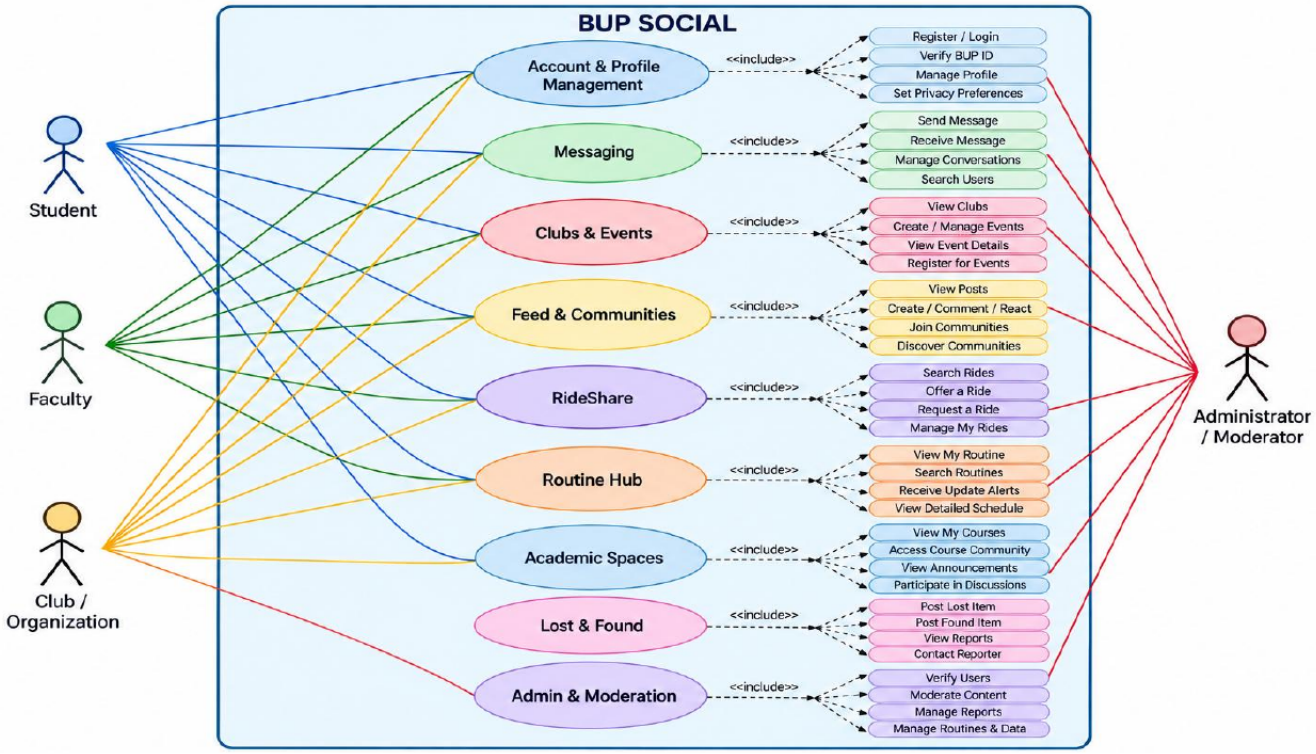
Figure: Context Diagram of BUP Social

Use Case Diagram — BUP Social

Four primary actors interact with the BUP Social platform: Student, Faculty Member, Club/Organization Admin, and System Administrator. Students can register and log in, manage their profiles, access the personalized feed, join communities, create and interact with posts, communicate through messaging, view academic information and routines, participate in clubs and events, use RideShare and Travel Partner services, access Lost & Found, and submit reports when necessary. Faculty members can manage their profiles, access academic spaces, provide or view relevant academic and routine information, and participate in clubs and events. Club/Organization Admins can manage their club profiles, publish club activities and events, and manage related community content. The System Administrator manages user verification, platform administration, routine information, reports, and content moderation.

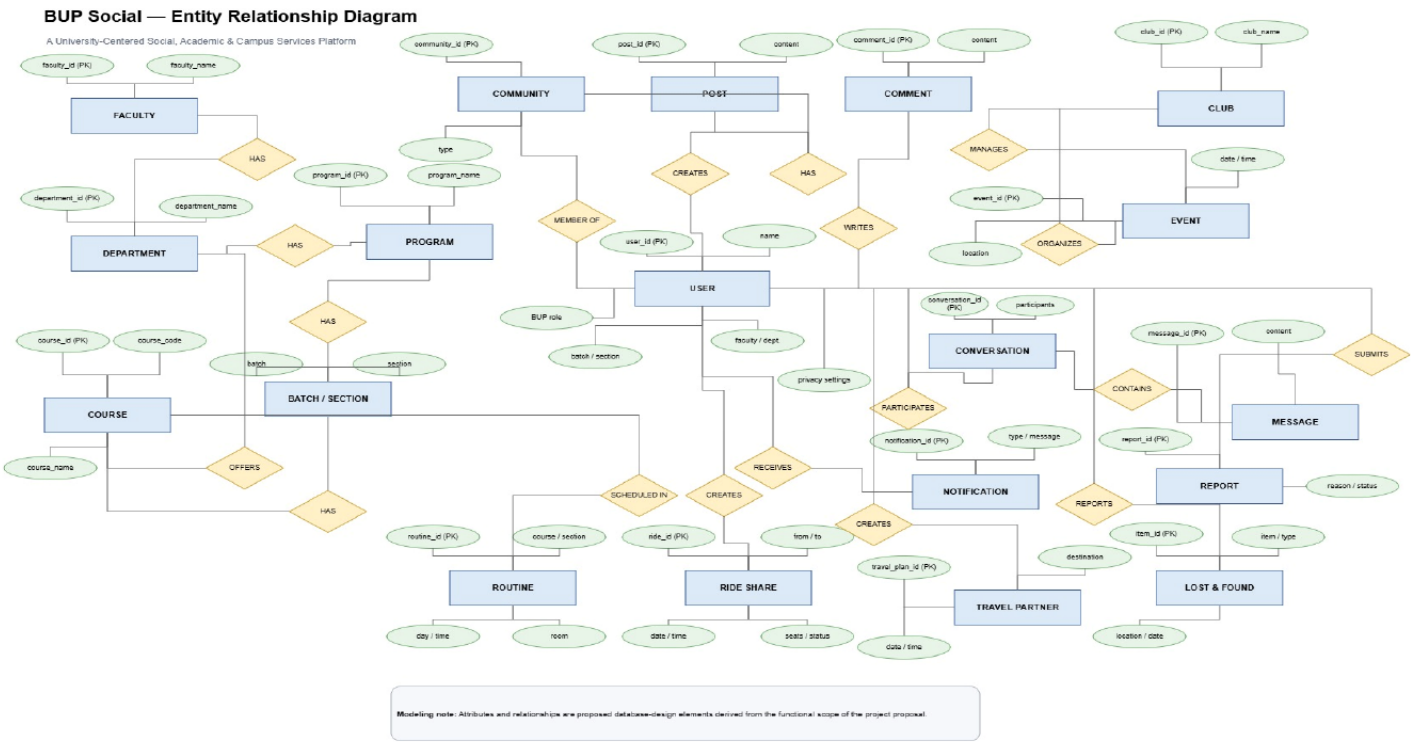
Several use cases are connected through <<include>> relationships where a function is required as part of another operation. For example, accessing personalized services includes User Authentication/Verification, while creating or managing content may include Authorization and Validation. <<extend>> relationships can represent optional or conditional actions, such as reporting content, sending notifications, or performing moderation when a reported activity requires administrative attention. All actors interact with the central BUP Social system according to their assigned roles and permissions.

7.5 Use-Case Overview



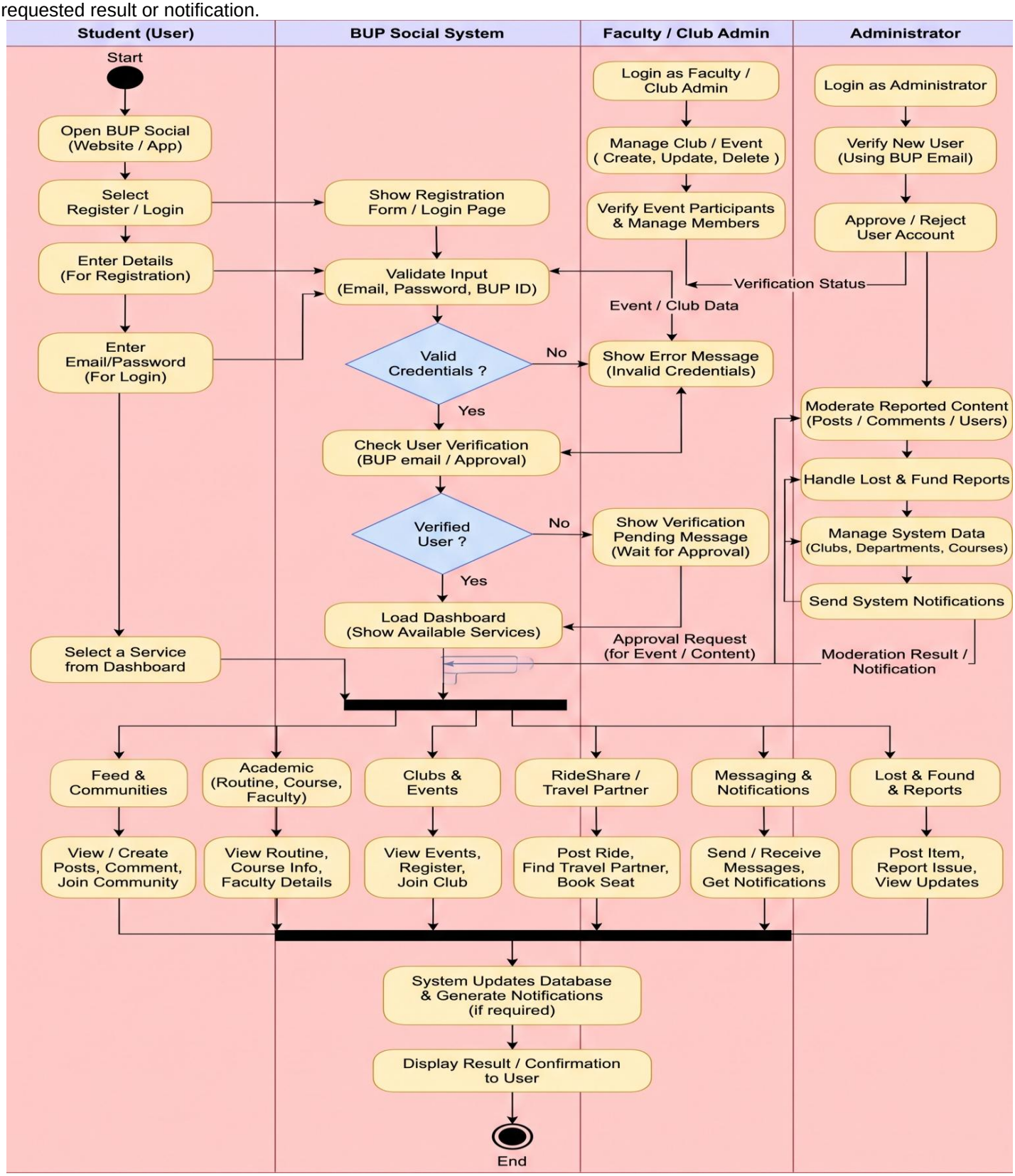
Entity-Relationship Diagram (ERD)

The BUP Social data model centers on a USER entity connected to academic structures such as FACULTY, DEPARTMENT, PROGRAM, BATCH/SECTION, and COURSE. Users create posts and comments, participate in communities and conversations, receive notifications, and interact with clubs and events. The model also represents routines, RideShare records, Travel Partner records, Lost & Found records, and moderation reports. These relationships support role-aware access, traceable records, and the platform's social, academic, mobility, and governance modules.



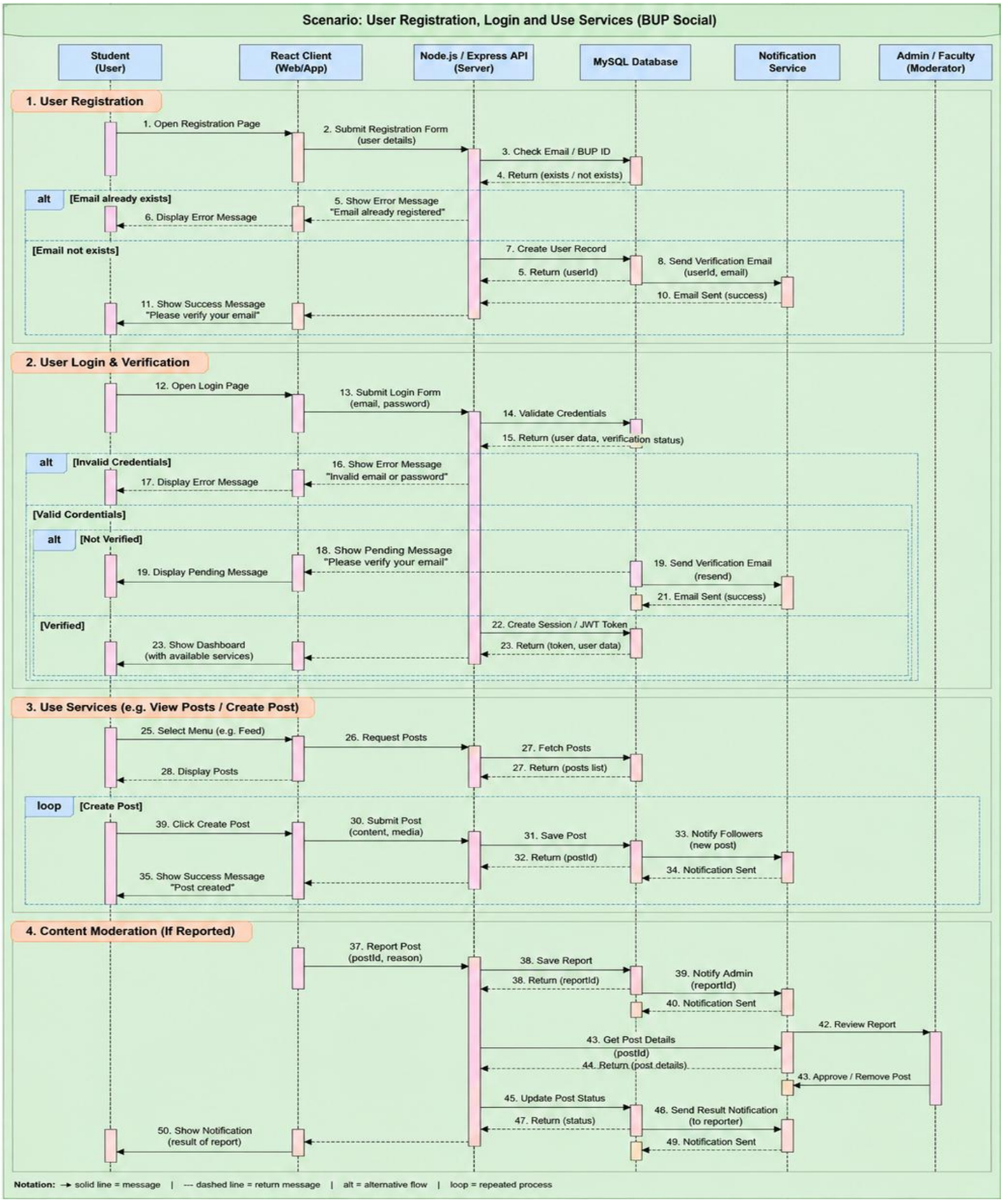
Activity Diagram

The diagram shows the **complete BUP Social user workflow** across four swimlanes: **Student, BUP Social System, Faculty/Club Admin, and Administrator**. It includes registration and login, account verification with retry for invalid credentials, dashboard access, and service selection. The workflow then branches into **Feed & Communities, Academic & Routine, Clubs & Events, RideShare/Travel Partner, Messaging & Notifications, and Lost & Found/Reports**, with faculty and club admins providing relevant information and administrators handling user verification, routine management, reports, and moderation. Finally, the system stores and retrieves the required data and provides the user with the requested result or notification.



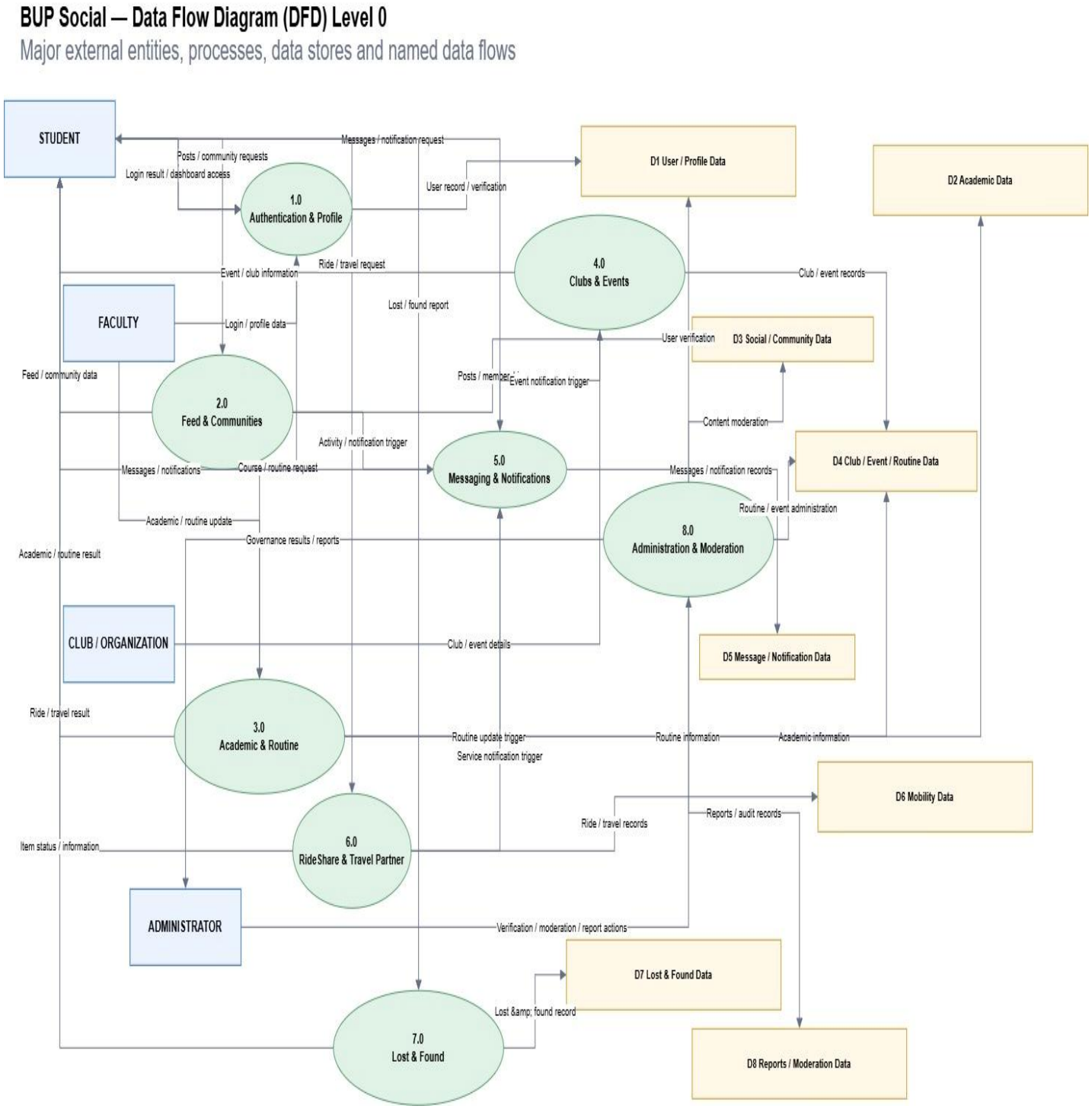
Sequence Diagram

The diagram illustrates the message flow between the Student, React web app, Node.js/Express server, MySQL database, and notification service during BUP Social activities. It covers the process from login and authentication to dashboard access and service requests, including Feed & Communities, Academic & Routine, Clubs & Events, Messaging, RideShare/Travel Partner, and Lost & Found. It also shows data validation, database queries and updates, service responses, and notification flows for relevant activities.



DFDLevel 0

This diagram breaks the BUP Social system into eight processes — authentication and profile management, feed and communities, academic and routine management, clubs and events, messaging and notifications, RideShare and Travel Partner, Lost & Found, and administration and moderation — and shows how they exchange data with eight internal data stores and four external parties: the student, faculty member, club/organization, and administrator.



7.5 Detailed Workflow

The workflow below represents a typical student interaction from authentication to a selected service, with common notification/record handling and a separate administration layer.

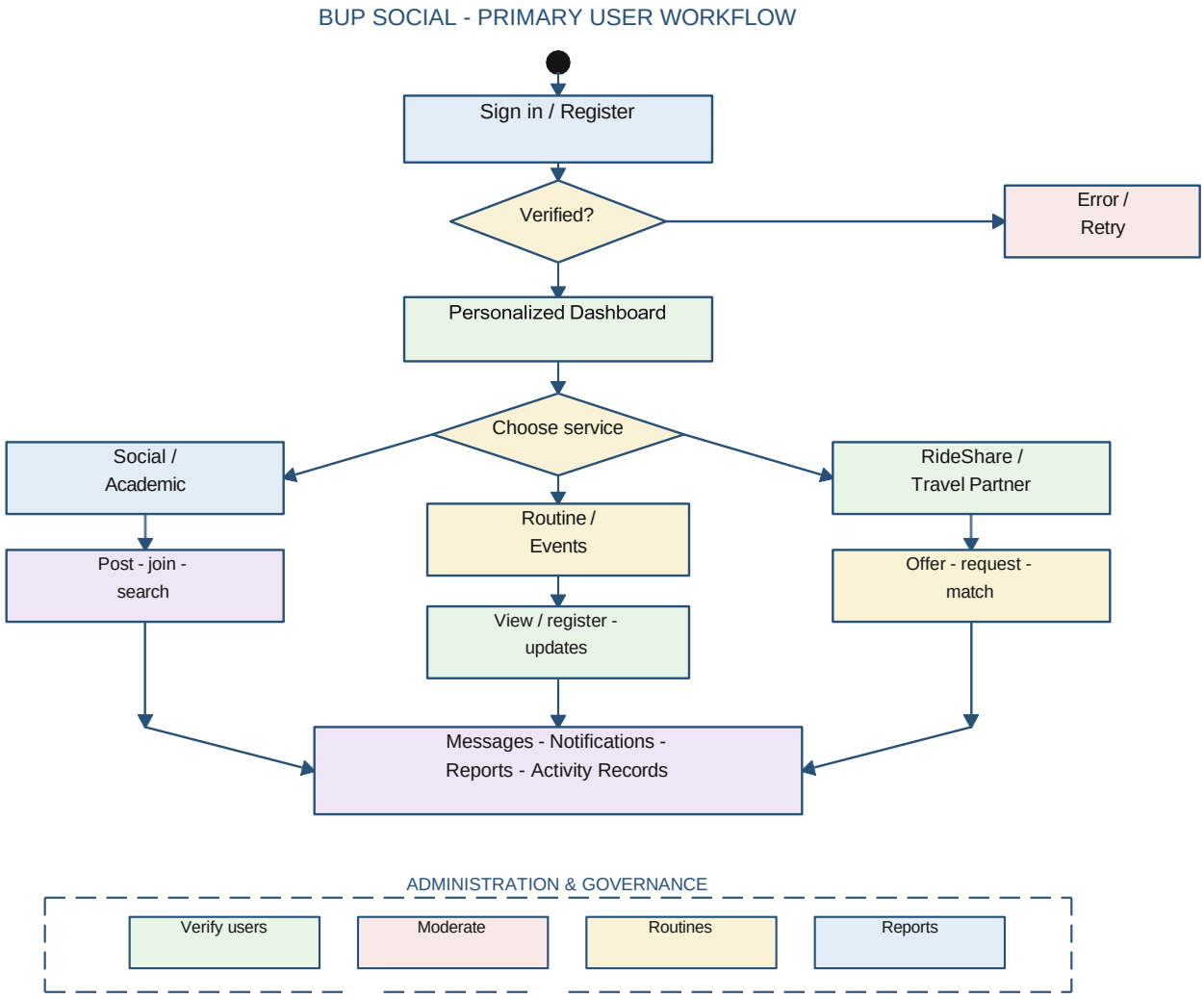


Figure 7.4: Representative BUP Social workflow

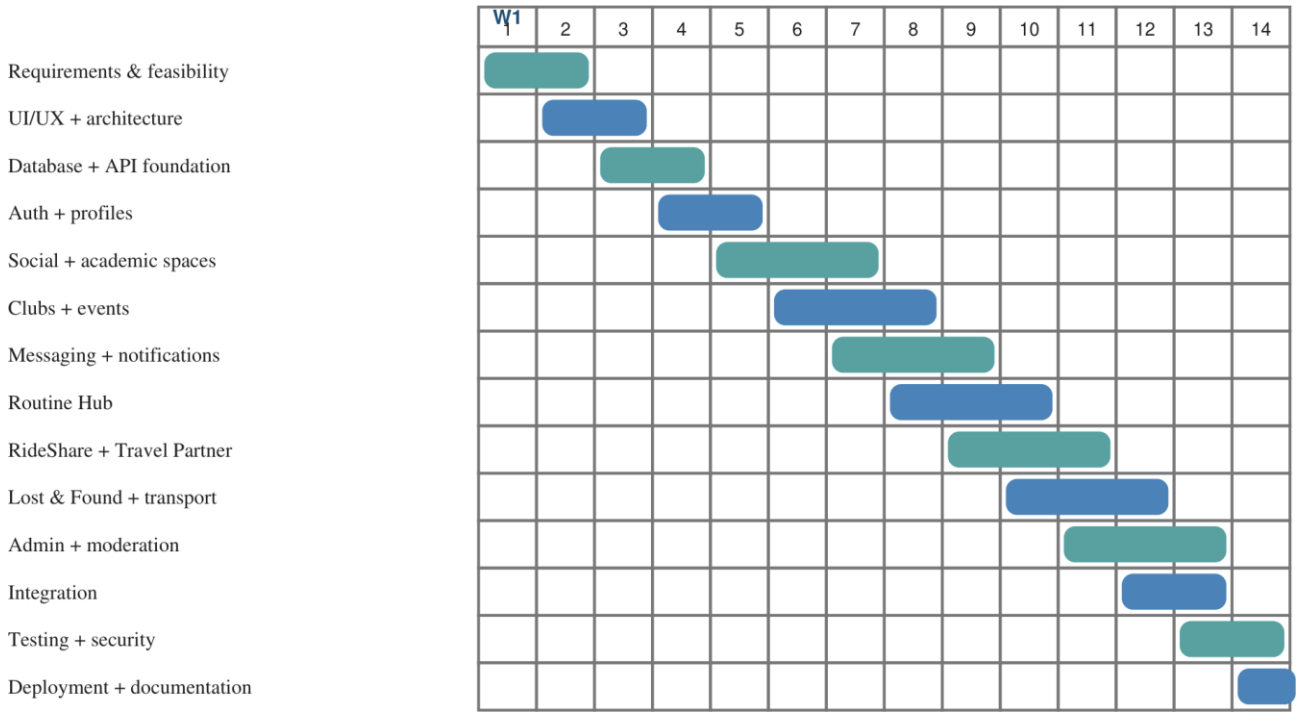
Access & Verification	Service Selection
Sign-in, registration and verification precede access to the personalized dashboard.	Students select Social/Academic, Routine/Events, or RideShare/Travel Partner services.
Shared Records & Notifications	**Administration & Governance**
Messages, notifications, reports and activity records provide a common record layer.	Verification, moderation, routine handling and reporting remain under administrative control.

8 Project Timeline

The project is planned over 14 weeks. Activities overlap where appropriate so that integration and testing begin before the final implementation week.

Work Package	Target Weeks
Requirements, feasibility & backlog	1–2
UI/UX prototype & system modeling	2–3
Database schema & API foundation	3–4
Authentication, roles & profiles	4–5
Social feed & academic communities	5–7
Clubs, events & notifications	6–8
Messaging	7–9
Routine Hub	8–10
RideShare & Travel Partner	9–11
Transport info & Lost & Found	10–12
Administration & moderation	11–13
Integration & regression testing	12–13
Security/usability testing & fixes	13–14
Deployment, documentation & final demo	14

Figure 8.1: 14-week implementation roadmap



9 Expected Outcome

The project is expected to produce a functional, mobile-responsive BUP Social prototype that demonstrates a coherent university community platform rather than a collection of disconnected features. A student should be able to authenticate, configure an academic profile, participate in relevant communities, view routines, discover events, communicate with other users, and use transportation-oriented services from the same environment.

The prototype should also provide verified clubs and organizations with structured publishing capabilities and give administrators centralized tools for user verification, moderation, routine management, reporting, and oversight. From a software engineering perspective, the project will demonstrate requirements engineering, system modeling, database design, API development, authentication, authorization, testing, version control, and deployment.

Expected Benefits

- Students: faster discovery of relevant academic, social, event and transportation information.
- Faculty: more structured communication with appropriate academic communities.
- Clubs: dedicated visibility, event promotion and member engagement.
- Administration: centralized governance, moderation, routine management and reporting.

10 Conclusion

BUP Social addresses a practical campus problem: important student information and coordination activities are currently distributed across platforms that were not designed around BUP's academic structure. The proposed system brings the most relevant activities into one role-aware web platform while keeping the first implementation within a realistic academic scope.

The proposal prioritizes a maintainable architecture, clear requirements, secure access control, privacy-aware transportation features, and incremental development. Future versions can extend the prototype with institutional integrations, native mobile applications, live transport tracking, analytics, and intelligent recommendations without changing the fundamental platform model.

References

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